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 Studierichting: Informatica
 Oefeningenreeks 7B nr. 2.1

$$w = \ln(x^2 + y^2 + z^2), \text{ met } x = s - t, y = s + t, z = 2\sqrt{st}$$

Stap 1: Afgeleiden van w

$$\frac{\partial w}{\partial x} = \frac{2x}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial y} = \frac{2y}{x^2 + y^2 + z^2}$$

$$\frac{\partial w}{\partial z} = \frac{2z}{x^2 + y^2 + z^2}$$

Stap 2: Afgeleiden van x , y en z

$$x = s - t \Rightarrow \frac{\partial x}{\partial s} = 1, \frac{\partial x}{\partial t} = -1$$

$$y = s + t \Rightarrow \frac{\partial y}{\partial s} = 1, \frac{\partial y}{\partial t} = 1$$

$$z = 2\sqrt{st} \Rightarrow \frac{\partial z}{\partial s} = \frac{t}{\sqrt{st}}, \frac{\partial z}{\partial t} = \frac{s}{\sqrt{st}}$$

Stap 3: Toepassen van de kettingregel voor $\frac{\partial w}{\partial s}$:

$$\begin{aligned} \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial s} \\ &= \frac{2x}{x^2 + y^2 + z^2} \cdot 1 + \frac{2y}{x^2 + y^2 + z^2} \cdot 1 + \frac{2z}{x^2 + y^2 + z^2} \cdot \frac{t}{\sqrt{st}} \\ &= \frac{2x + 2y + 2z \cdot \frac{t}{\sqrt{st}}}{x^2 + y^2 + z^2} \end{aligned}$$

Stap 4: Toepassen van de kettingregel voor $\frac{\partial w}{\partial t}$:

$$\begin{aligned} \frac{\partial w}{\partial t} &= \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial t} + \frac{\partial w}{\partial z} \cdot \frac{\partial z}{\partial t} \\ &= \frac{2x}{x^2 + y^2 + z^2} \cdot (-1) + \frac{2y}{x^2 + y^2 + z^2} \cdot 1 + \frac{2z}{x^2 + y^2 + z^2} \cdot \frac{s}{\sqrt{st}} \\ &= \frac{-2x + 2y + 2z \cdot \frac{s}{\sqrt{st}}}{x^2 + y^2 + z^2} \end{aligned}$$

Stap 5: Invullen van x , y en z :

Teller van $\frac{\partial w}{\partial s}$:

$$2x + 2y + 2z \cdot \frac{t}{\sqrt{st}} = 2(s-t) + 2(s+t) + 2 \cdot 2\sqrt{st} \cdot \frac{t}{\sqrt{st}}$$

$$= 4s + 4t$$

Teller van $\frac{\partial w}{\partial t}$:

$$-2x + 2y + 2z \cdot \frac{s}{\sqrt{st}} = -2(s-t) + 2(s+t) + 2 \cdot 2\sqrt{st} \cdot \frac{s}{\sqrt{st}}$$

$$= 4s + 4t$$

Noemer:

$$x^2 + y^2 + z^2 = (s-t)^2 + (s+t)^2 + (2\sqrt{st})^2$$

$$= 2s^2 + 4st + 2t^2 = 2(s+t)^2$$

Stap 6: Einduitkomst:

$$\frac{\partial w}{\partial s} = \frac{\partial w}{\partial t} = \frac{4s + 4t}{2(s+t)^2} = \frac{2(s+t)}{(s+t)^2} = \frac{2}{s+t}$$

References